Alexander Kim

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Education

Duke University, Durham, NC — Aug 2026–May 2031
Doctor of Philosophy (PhD), Electrical and Computer Engineering
Advisor: Dr. Rishikesan Kamaleswaran
Honors and Awards: 2026 Pratt-Gardner Fellow

University of Wisconsin–Madison, Madison, WI — Jan 2026–May 2026
Non-degree coursework (University Special student)
Coursework: Measure Theory; Theory and Algorithms for Data Science; Data Structures and Algorithms

Hamilton College, Clinton, NY — Aug 2021–May 2025
Bachelor of Arts (BA), Mathematics (Minor: Music) — GPA: 3.5 / 4.0
Selected Coursework: Deep Learning; Real Analysis; Graph Theory; Modern Algebra; Linear Optimization; Statistical Modeling & Applications; Electronics and Computers; Design Principles
Honors and Awards: Emerson Summer Research Fellow; Dean's List

Research Experience

- **Emerson Summer Research Fellow** — May 2024–Aug 2024
  Hamilton College, Clinton NY — *Advisor: Prof. Charlotte Botha, PhD*
  During my Emerson Fellowship, I developed **AVRA**, a real-time inference system for vocal register analysis that integrates audio to image processing, supervised learning models, and an interactive interface for pedagogical use.
  Signal Processing and Data Pipeline. Engineered an automated audio–spectrogram pipeline that segments vocal tracks into 3-second units, synthesizes high-resolution Mel-spectrograms, and indexes segment metadata via a SQLite/SQLAlchemy backend. Implemented spectrogram tiling (128×154) for standardized ML input across 14,000+ samples.
  Model Development and Sliding-Window Inference. Trained a custom PyTorch CNN (96% accuracy) and RBF-SVM (94% accuracy) for four-class register classification. Designed a sliding-window inference algorithm to detect register transitions, compute temporal stability metrics, and estimate register-frequency distributions over continuous audio.
  Full-Stack Real-Time Application. Built a FastAPI + React system enabling users to upload audio, interactively select waveform regions, toggle CNN/SVM inference, and receive annotated spectrograms with transition markers in < 150 ms. Created a visualization dashboard (Chart.js) to display register distributions and stability analytics. Docker container available to Hamilton College users.
  Research Output. First-author preprint submitted to *The Journal of New Music Research* and is currently under review.

- **Research Intern** — Jun 2022–Aug 2023
  Boston Children's Hospital / Harvard Medical School — *PI: Dr. Ata Kiapour, PhD*
  As an undergraduate researcher in the Musculoskeletal Informatics Group, I worked across 3D surgical simulation, clinical NLP, and imaging pipelines to support diagnostic and pre-operative decision-making in pediatric orthopedics.
  VirtualHip Mesh-Repair and Surgical Simulation Algorithms. Implemented subject-specific 3D surgical planning algorithms for periacetabular osteotomy (3-cut/4-cut PAO) and femoral osteotomy (VDRO/VDO) using Python, `trimesh`, `scipy`, and `pymeshlab`. Encoded femoral neck–shaft angle correction as a target-driven optimization and processed cases into clinically meaningful fragment meshes. Developed watertight-repair and boundary-capping routines plus collision-based fragment identification to robustly separate acetabular and pelvic fragments. Automated the derivation of osteotomy planes from anatomical landmarks via best-fit planes and spherical/cylindrical acetabulum approximations, then used iterative search over plane rotations/translations to satisfy geometric and structural constraints. These algorithms supported VirtualHip's reduction of pre-operative planning time by $\sim$20%.
  Clinical NLP, Domain Adaptation, and Multimodal Cohorts. Worked on an orthopedic diagnosis and outcomes dataset by processing and annotating 2,500+ ACL-related EHR notes spanning a 20-year period. Contributed to the InterDAPT framework for domain-adaptive clinical NER: pretraining

BioClinicalBERT on BCH clinical corpora (BCH-BERT), applying weakly supervised NER fine-tuning on machine-labeled operative notes (InterNER-BERT), and then supervising final fine-tuning on our manually annotated orthopedic notes. This pipeline supported automatic extraction of diagnoses, ligament/meniscal injuries, and surgical details used in downstream analyses. Co-authored resulting publications in *Am. J. Sports Med. 2023* and *ACL Anthology BioNLP 2023*.

Imaging Segmentation Pipelines and VirtualHip Training Data. Prepared, and corrected imaging data for 100+ MRI and CT studies of pelvis and hip anatomy. Generated standardized mesh and label inputs for the VirtualHip segmentation pipeline, including region-wise element indexing from clinical models and post-processing of segmentation outputs. This work improved robustness and usability of the VirtualHip workflow and provided training/evaluation data for subsequent segmentation and simulation models in pediatric hip dysplasia and related pathologies.

Research Focus Areas

Representation Learning and Geometry, Clinical AI, Multi-agent interoperability

Peer-Reviewed Publications

1. Williamson, A. *et al.* **Kim, A.** (2026). Large Language Models for Zero-Shot Procedure Extraction in Orthopedic Surgery: A Comparative Evaluation. `https://www.nature.com/articles/s41598-026-52928-3` *scientific reports*.
2. Suresh, S. *et al.* (2023) **Kim, A.** (2023). Intermediate Domain Finetuning for Weakly Supervised Domain-Adaptive Clinical NER. *ACL Anthology BioNLP*. `https://aclanthology.org/2023.bionlp-1.29/`
3. Pruneski, J. *et al.* **Kim, A.** (2023). Concomitant Meniscal / Ligament Injuries Associated with Pediatric ACL Surgery. *Am. J. Sports Med.* `https://journals.sagepub.com/doi/full/10.1177/03635465231205556`
4. Salt, M. D., **Kim, A. (second author)**, Flaherty, M. R., *et al.* (2020). Distracted-driving Laws and Teenage Crash Fatalities. *Pediatrics* 145(6). `https://doi.org/10.1542/peds.2019-3621`

Pre-print

1. **Kim, A.**, Botha, C. (2024). Machine-Learning Approaches to Vocal-Register Classification in Contemporary Male Pop Music. *arXiv*. `https://arxiv.org/abs/2505.11378`

Software Projects

SOFIA — Offline LLM Agent Framework (22 GitHub Stars) Feb 2025–Jul 2025

- Developed a local LLM agent system integrating Ollama models with a middleware orchestration layer for tool execution, CLI automation, and multi-agent workflows.
- Implemented OmniParser-based OCR and GUI-control pipelines enabling screen parsing, clickable element detection, and desktop-automation actions.
- Built MCP-compatible tools for Gmail and Google Calendar, allowing structured email retrieval, scheduling, and reasoning from local LLMs without cloud dependency.

tensorkit-learn — Custom ML Library (C++/Python) Jan 2025–Jul 2025

- Implemented a tensor library in C++ with Python bindings, supporting 1D/2D tensors, broadcasting, reductions, and differentiable arithmetic to match ML workflows.
- Built from-scratch implementations of SVM, GLM (iterative solver with convergence criteria), and MLP modules, including custom loss functions and C++ link functions.
- Designed a PyTorch-style dataloader class, supporting batching and shuffling of tabular datasets.
- Created 32 unit tests and reproducible installation scripts (Bash) enabling reliable compilation and deployment across Linux and macOS.

Industry Experience

- **Epic Systems**, Madison, WI
  Project Manager — Revenue Cycle (Aug 2025 – Dec 2025)
  – Completed certification tracks in Resolute Hospital Billing (Administration and Charging), enabling configuration, testing, and deployment of enterprise-scale billing workflows.
  – Led implementation workgroups with hospital partners, conducting requirements gathering, build design, regression testing, and end-user validation for production go-lives.
  – Coordinated cross-functional technical teams (billing, claims, behavioral-health, reporting) to ensure

system reliability, integration correctness, and regulatory compliance.
  - Configured 100+ charge codes and 10+ Revenue Guardian checks, collaborating with hospital IT and Revenue Integrity teams to design triggers and modifier rules during onsite support trips.
- **tmc**, Boston, MA
  ML Software Engineer (Mar 2025 – Jul 2025)
  - Developed a cross-platform invoicing MVP in React Native and TypeScript, including UI state management, camera/scan interface, and offline-first data handling.
  - Built an OCR pipeline using HuggingFace models and deployed it as a scalable Dockerized Sanic REST service with batching, and async I/O.
  - Worked on receipt-parsing logic and a structured extraction layer feeding into a graph database (Memgraph) and parallel financial-transaction ledger (TigerBeetle).
  - Created Jupyter-based evaluation notebooks for OCR accuracy benchmarking, and workflow testing.

Technical Skills

Languages: Python, C/C++, TypeScript, Java, R, SQL, Go, LaTeX, Lean
Libraries: PyTorch, scikit-learn, pandas, NumPy, OpenCV, Matplotlib
Frameworks: Svelte, FastAPI, Sanic, MCP, CUDA
Tools: Vim, Git, Docker, Linux, Bun, Google Cloud Platform

References

- **Dr. Ata Kiapour PhD**
  Principal Investigator, Musculoskeletal Informatics Group
  Boston Children's Hospital / Harvard Medical School
  `Ata.Kiapour@childrens.harvard.edu`
  (617) 355-6000
- **Professor Sally Cockburn PhD**
  Mathematics Academic Advisor
  Samuel F. Pratt Professor of Mathematics & Statistics
  `scockbur@hamilton.edu`
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- **Professor Saber Ahmed PhD**
  Assistant Professor of Mathematics & Statistics
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- **Professor Shawn Chen PhD**
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- **Professor Charlotte Botha PhD**
  Emerson Research Project Advisor
  Assistant Professor of Music
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